LEARNER GUIDE

Where two lines meet

Equations, Inequalities and Simultaneous Relationships · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A single line is a whole set of solutions. Take **two** lines and ask: which pair (x, y) satisfies *both* equations at once? Since each line is its own equation's solution set, a pair on **both** lines must be where they **cross**. That crossing is the **common solution** — the answer to the two equations taken together (solved **simultaneously**). You can read it off the graph and confirm it by substituting into each equation: the common solution is the one point that makes both true.

What you are learning to notice

- The point where two lines cross lies on **both** lines.
- It is the **common solution**: it satisfies both equations at once.
- Points on just one line satisfy only that equation — not both.
- Check the crossing by substituting into *each* equation; both must come out true.

Moving through it, one step at a time

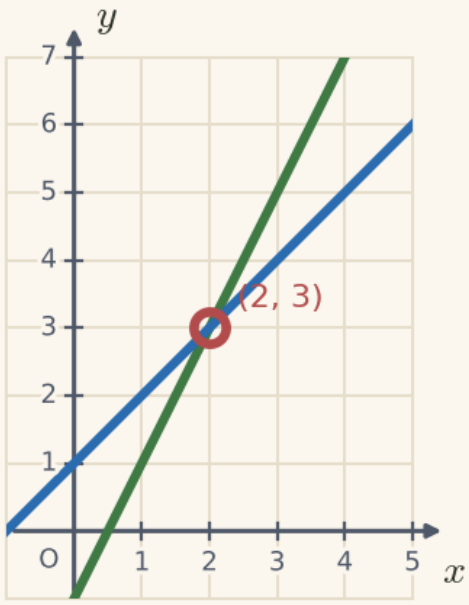
Plot both lines on the same axes and read the point where they cross. To be sure, substitute that point into each equation in turn — it must satisfy both. Algebraically, the crossing is found by setting the two right-hand sides equal (because at the crossing the y-values agree), solving for x, then finding y. Either way, one pair answers both equations together.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Reading the common solution

Where do $y = 2x - 1$ and $y = x + 1$ cross?

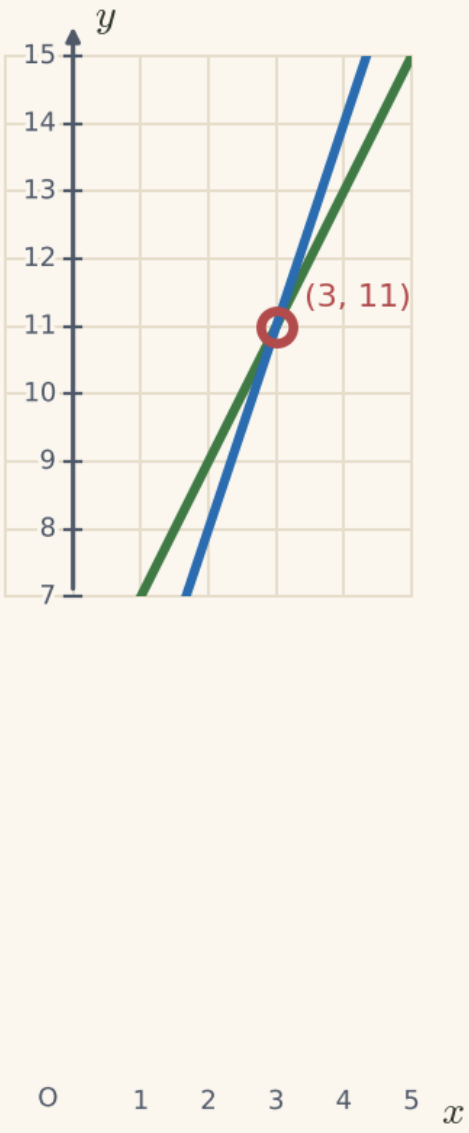


1. At the crossing the y-values agree, so set the sides equal.	$2x - 1 = x + 1$
2. Collect the x-terms and solve.	$x = 2$
3. Find y from either equation.	$y = 2 + 1 = 3$
4. Check in both.	$2\times2 - 1 = 3$ ✓ and $2 + 1 = 3$ ✓

(2, 3) — the crossing satisfies both equations, so it is the common solution.

Example 2 — A break-even point

Plan A costs $y = 2x + 5$; Plan B costs $y = 3x + 2$. When do they cost the same?



1. Set the two costs equal.	$2x + 5 = 3x + 2$
2. Solve for x (the hours).	$x = 3$
3. Find the shared cost.	$y = 2\times3 + 5 = 11$
4. Interpret the decision.	Same at (3, £11); cheaper plan differs either side

They break even at (3, £11) — the common solution. Before 3 hours one plan is cheaper; after, the other.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading a line as a set of solutions (Lesson 5), reading coordinates, and substituting a point), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.